

Numpy

```
In [2]: list=[1,2,4,5]
list
```

```
Out[2]: [1, 2, 4, 5]
```

```
In [10]: import numpy as np
```

```
In [12]: arr=np.array(list)
```

```
In [14]: arr
```

```
Out[14]: array([1, 2, 4, 5])
```

```
In [17]: m=np.random.randint(1,10,(4,6))      # 1-9
m                                              # 4 row , 6 column
```

```
Out[17]: array([[1, 6, 4, 1, 2, 5],
                [5, 4, 1, 9, 3, 3],
                [9, 9, 8, 6, 1, 5],
                [5, 9, 1, 8, 7, 8]], dtype=int32)
```

```
In [18]: print(m)      # 4*6
```

```
[[1 6 4 1 2 5]
 [5 4 1 9 3 3]
 [9 9 8 6 1 5]
 [5 9 1 8 7 8]]
```

```
In [19]: m[0]      # 0th row indexing
```

```
Out[19]: array([1, 6, 4, 1, 2, 5], dtype=int32)
```

```
In [20]: m[-3]     # -3 th row
```

```
Out[20]: array([5, 4, 1, 9, 3, 3], dtype=int32)
```

```
In [21]: m[2:5]     # 2th - 4th row slicing
```

```
Out[21]: array([[9, 9, 8, 6, 1, 5],
                [5, 9, 1, 8, 7, 8]], dtype=int32)
```

```
In [22]: m
```

```
Out[22]: array([[1, 6, 4, 1, 2, 5],
                [5, 4, 1, 9, 3, 3],
                [9, 9, 8, 6, 1, 5],
                [5, 9, 1, 8, 7, 8]], dtype=int32)
```

```
In [23]: m[ : ]     # all
```

```
Out[23]: array([[1, 6, 4, 1, 2, 5],
                [5, 4, 1, 9, 3, 3],
                [9, 9, 8, 6, 1, 5],
                [5, 9, 1, 8, 7, 8]], dtype=int32)
```

```
In [24]: len(m)      # no. of rows
```

```
Out[24]: 4
```

```
In [25]: m[0:3]
```

```
Out[25]: array([[1, 6, 4, 1, 2, 5],
                [5, 4, 1, 9, 3, 3],
                [9, 9, 8, 6, 1, 5]], dtype=int32)
```

```
In [26]: m
```

```
Out[26]: array([[1, 6, 4, 1, 2, 5],
                [5, 4, 1, 9, 3, 3],
                [9, 9, 8, 6, 1, 5],
                [5, 9, 1, 8, 7, 8]], dtype=int32)
```

column

```
In [27]: m
```

```
Out[27]: array([[1, 6, 4, 1, 2, 5],
                [5, 4, 1, 9, 3, 3],
                [9, 9, 8, 6, 1, 5],
                [5, 9, 1, 8, 7, 8]], dtype=int32)
```

```
In [28]: m[:,0]      # 0th column
```

```
Out[28]: array([1, 5, 9, 5], dtype=int32)
```

```
In [29]: m[:, -3]    # -3th column
```

```
Out[29]: array([1, 9, 6, 8], dtype=int32)
```

array

```
In [30]: m
```

```
Out[30]: array([[1, 6, 4, 1, 2, 5],
                [5, 4, 1, 9, 3, 3],
                [9, 9, 8, 6, 1, 5],
                [5, 9, 1, 8, 7, 8]], dtype=int32)
```

```
In [31]: m[2,3]      # 2nd row , 3rd column
```

```
Out[31]: np.int32(6)
```

```
In [32]: m[0, -3]    # 0th row , -3th column
```

```
Out[32]: np.int32(1)
```

```
In [33]: m[1:3 , 2:5]      # 1-2th row , 2-4th column
```

```
Out[33]: array([[1, 9, 3],
               [8, 6, 1]], dtype=int32)
```

```
In [34]: m
```

```
Out[34]: array([[1, 6, 4, 1, 2, 5],
               [5, 4, 1, 9, 3, 3],
               [9, 9, 8, 6, 1, 5],
               [5, 9, 1, 8, 7, 8]], dtype=int32)
```

shape , reshape

```
In [35]: m.shape      # total no. 24
```

```
Out[35]: (4, 6)
```

```
In [39]: m.reshape(6,4)      # so reshape also is of 24 no.
```

```
Out[39]: array([[1, 6, 4, 1],
               [2, 5, 5, 4],
               [1, 9, 3, 3],
               [9, 9, 8, 6],
               [1, 5, 5, 9],
               [1, 8, 7, 8]], dtype=int32)
```

masking/filter

```
In [40]: m
```

```
Out[40]: array([[1, 6, 4, 1, 2, 5],
               [5, 4, 1, 9, 3, 3],
               [9, 9, 8, 6, 1, 5],
               [5, 9, 1, 8, 7, 8]], dtype=int32)
```

```
In [41]: m<3      # condition - ans in True,False
```

```
Out[41]: array([[ True, False, False,  True,  True, False],
               [False, False,  True, False, False, False],
               [False, False, False, False,  True, False],
               [False, False,  True, False, False, False]])
```

```
In [42]: m[m<5]
```

```
Out[42]: array([1, 4, 1, 2, 4, 1, 3, 3, 1, 1], dtype=int32)
```

```
In [43]: m[m!=2]
```

```
Out[43]: array([1, 6, 4, 1, 5, 5, 4, 1, 9, 3, 3, 9, 9, 8, 6, 1, 5, 5, 9, 1, 8, 7,
               8], dtype=int32)
```

convert image into array

image analysis using np,PIL

```
In [47]: import matplotlib.pyplot as plt    #matplotlib for visualization (graph)
```

```
In [48]: from PIL import Image    # Python Imaging Libraby
```

```
In [54]: tiger_img=Image.open(r'C:\Users\LENOVO\Downloads\tiger_img.jpg')
```

```
In [55]: tiger_img
```

Out[55]:



```
In [56]: type(tiger_img)
```

Out[56]: PIL.JpegImagePlugin.JpegImageFile

```
In [58]: tiger_arr=np.asarray(tiger_img)  
tiger_arr
```

```

Out[58]: array([[123, 71, 23],
               [124, 72, 24],
               [125, 73, 25],
               ...,
               [246, 177, 60],
               [246, 177, 60],
               [246, 177, 60]],

             [[121, 71, 22],
              [123, 71, 23],
              [124, 72, 24],
              ...,
              [246, 177, 60],
              [246, 177, 60],
              [246, 177, 60]],

             [[120, 69, 22],
              [121, 71, 22],
              [122, 72, 23],
              ...,
              [246, 177, 60],
              [246, 177, 60],
              [246, 177, 60]],

             ...,

             [[ 93, 58, 38],
              [ 87, 52, 32],
              [ 85, 52, 33],
              ...,
              [137, 75, 36],
              [135, 73, 34],
              [134, 72, 33]],

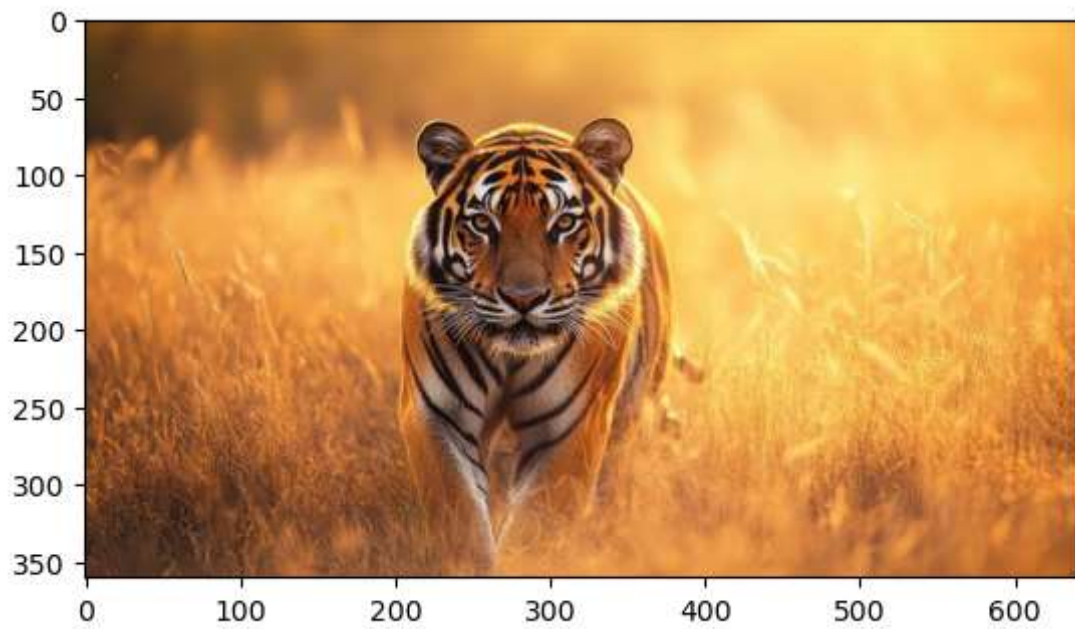
             [[101, 66, 44],
              [ 94, 59, 39],
              [ 89, 57, 36],
              ...,
              [135, 73, 36],
              [134, 72, 35],
              [132, 70, 33]],

             [[ 98, 63, 41],
              [101, 66, 44],
              [ 96, 61, 41],
              ...,
              [132, 72, 36],
              [131, 71, 35],
              [129, 69, 33]]], shape=(360, 643, 3), dtype=uint8)

```

```
In [59]: plt.imshow(tiger_img)
```

```
Out[59]: <matplotlib.image.AxesImage at 0x2401eeca3c0>
```



```
In [60]: tiger_arr.shape
```

```
Out[60]: (360, 643, 3)
```

```
In [ ]:
```